

Sydney Girls High School



November 2014
MATHEMATICS
EXTENSION 1

YEAR 12
ASSESSMENT TASK 1 for HSC 2015

Time Allowed: 60 minutes
+ 5 minutes Reading Time

Topics: Mathematical Induction, Parametric Equations and Circle Geometry.

General Instructions:

- There are Seven (7) Questions of equal marks.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each Question on a new page.
- Write on one side of paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number clearly at the top of each question and clearly number each question.

Total: 56 marks

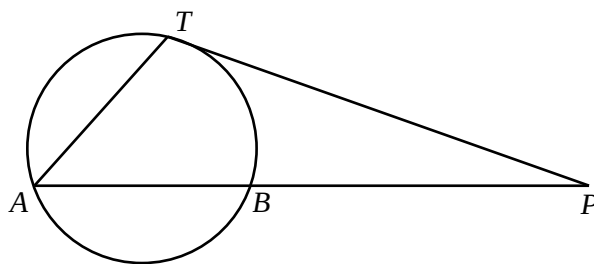
Student Name: _____ **Teacher Name:** _____

Question 1 (8 marks)

- a) Find the equation of the tangent to the parabola $x = 6t$, $y = 3t^2$ at the point where $t = 2$. 3
- b) Normals are drawn at two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ to the parabola $x^2 = 4ay$.
- i) Derive the equation of the normal at P . 2
- ii) Find the point of intersection of these two normals. 3
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Question 2 (8 marks)

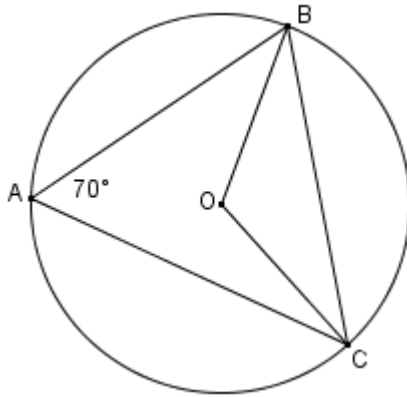
- a) Copy or trace the diagram onto your answer page.



- i) PT is a tangent to the circle. Prove that $\triangle TBP \sim \triangle ATP$. 2
- ii) Given that $TP = 12$ cm and $AP = 16$ cm, find BP . 2
- b) Prove by mathematical induction that $3^{4n} - 1$ is divisible by 80 for $n \geq 1$. 4

Question 3 (8 marks)

a)



- | | | |
|-----|---------------------------------|---|
| i) | Find $\angle COB$ with reasons. | 2 |
| ii) | Find $\angle OCB$ with reasons. | 2 |

- | | | |
|----|---|---|
| b) | Prove by Mathematical induction that $4^n \geq 1 + 3n$ for all $n \geq 1$. | 4 |
|----|---|---|
-

Question 4 (8 marks)

- | | | |
|--|--|---|
| a) | Derive the Cartesian equation for $x = 2t - 1, y = 3t^2 + 1$ | 2 |
| b) The chord PQ of the parabola $x^2 = 12y$ passes through the point $C(8, 0)$.
If $P(6p, 3p^2)$ and $Q(6q, 3q^2)$. | | |
| i) | Derive the equation of the chord PQ . | 2 |
| ii) | Show that $4(p + q) = 3pq$. | 1 |
| iii) | Find the locus of M , midpoint of PQ . | 3 |

Question 5 (8 marks)

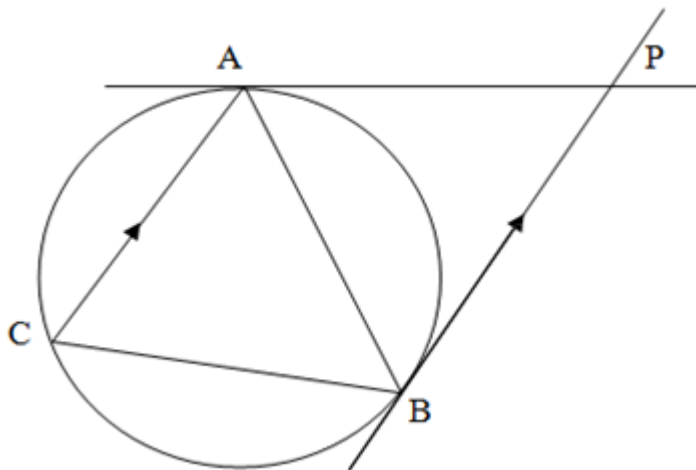
a) Consider the points A , B and C lying on the circle, with tangents drawn from A and B , meeting at P . AC is parallel to BP . Copy the diagram.

i) Let $\angle CAB = x^\circ$. Prove $AB = BC$.

2

ii) Prove $\angle ABC = \angle APB$.

2



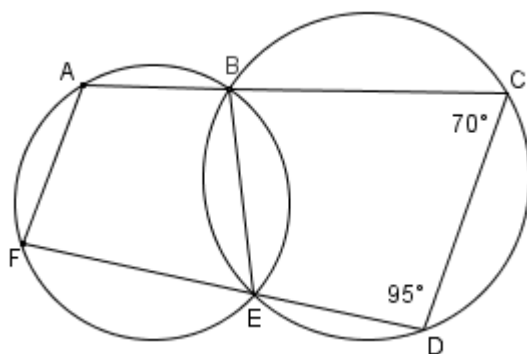
b) $P(2p, p^2)$ and $Q(2q, q^2)$ are two points on the parabola $x^2 = 4y$.

The chord PQ subtends a right angle at the origin O . Show that $pq = -4$.

2

c) Find $\angle AFE$, give reasons.

2



Question 6 (8 marks)

- a) Use Mathematical Induction to prove for all integers $n \geq 1$ 4
 $1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \cdots + n \times 2^n = (n-1)2^{n+1} + 2$
- b) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.
The tangent at P and the line through Q parallel to the y axis intersect at T .
- i) Draw a diagram to illustrate the information above. 1
- ii) Show that the coordinates of T are $(2aq, 2apq - ap^2)$, given the tangent at P is $y = px - ap^2$ 1
- iii) Find the locus of M , the midpoint of PT when $pq = -1$. 2

Question 7 (8 marks)

- a) A chord of contact to the parabola $x^2 = 4y$, has the equation $y = x + 3$.
Determine the external point from which the tangents are drawn.

2

- b) $ABCDE$ is a pentagon where $AB = AE$, $\angle AEB = x$ and $\angle BAC = y$.

- i) Prove that $\angle AEB = \angle ECA$

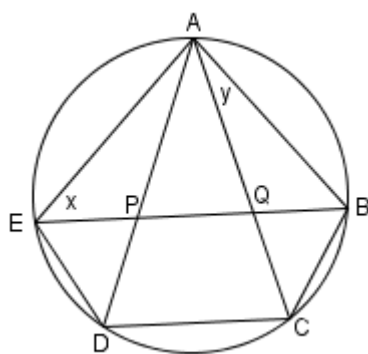
1

- ii) Prove that $\angle BQC = x + y$

2

- iii) Prove that $PQCD$ is a cyclic quadrilateral.

3



The End.

EXT 1 TASK 1 2014.

Q1

$$a) \quad t = \frac{x}{6}$$

$$y = 3 \left(\frac{x}{6} \right)^2$$

$$= \frac{3x^2}{36}$$

$$= \frac{x^2}{12}$$

$$y' = \frac{x}{6}$$

$$\text{at } t=2 \quad x=12$$

$$y' = 2$$

$$y - 12 = 2(x - 12)$$

$$y = 2x - 24 + 12$$

$$\boxed{y = 2x - 12}$$

bii)

$$x = ap^3 - yp + 2ap$$

$$ap^3 + yq + 2ap - yp = aq^3 + 2aq$$

$$y(q-p) = a(q^3 - p^3) + 2a(q-p)$$

$$y = a(q^2 + pq + p^2) + 2a$$

$$y = a(q^2 + pq + p^2 + 2)$$

$\therefore$

$$x = ap^3 - p(q^2 + pq + p^2 + 2) + 2ap$$

$$= ap^3 - pq^2 - p^2q - p^3 - 2p + 2ap$$

$$= -apq(p+q) \quad \checkmark$$

many students have poor setting out and didn't simplify their answer.

$$b)i) \quad x^2 = 4ay$$

$$m_2 = -\frac{1}{p}$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a}$$

$$\text{at } x = 2ap$$

$$m_1 = p$$

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

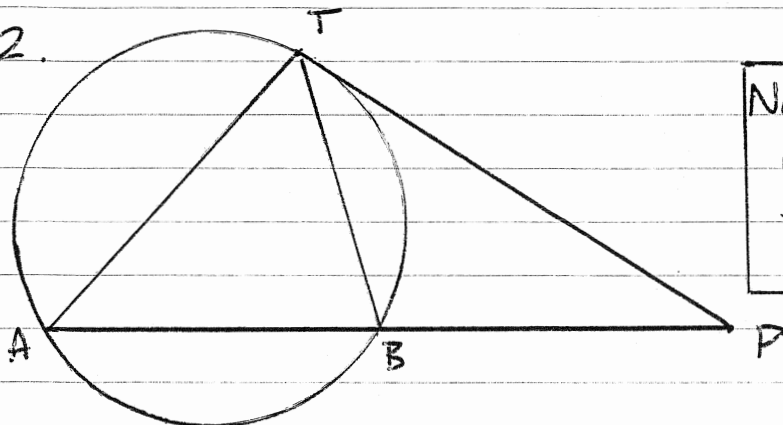
$$yp - ap^3 = -x + 2ap \quad \checkmark \checkmark$$

$$\boxed{x + yp = ap^3 + 2ap}$$

$$x + yq = aq^3 + 2aq$$

Q2.

a.



Notes: Part a. done well. Another solution for ii used $AP \cdot PB = PT^2$ and included angle P.

i. In $\triangle TBP$, $\triangle ATP$,
 $\angle PTB = \angle PAT$ ($\angle$ between tangent and chord
 $= \angle$ in alternate segment)
 $\angle BPT = \angle TPA$ ($\angle P$ is common)

$\therefore \triangle TBP \sim \triangle ATP$ (equiangular)

ii. $AP \cdot PB = PT^2$ (square of tangent = product of intercepts of secant)

$$\therefore 16 \times PB = 12^2$$

$$PB = \frac{144}{16} = 9 \text{ cm}$$

b. Prove that $3^{4n} - 1 = 80p$ where p is an integer
 for $n \geq 1$

step 1: show true for $n=1$

$$\begin{aligned} \text{LHS} &= 3^4 - 1 \\ &= 81 - 1 \\ &= 80 \\ &= \text{RHS with } p=1 \end{aligned}$$

step 2: assume true for $n=k$
 ie, $3^{4k} - 1 = 80q$

step 3: prove true for $n=k+1$

ie, prove that: $3^{4(k+1)} - 1 = 80p$

$$\begin{aligned} \text{LHS} &= 3^{4k+4} - 1 \\ &= 3^{4k} \times 3^4 - 1 \\ &= (80q + 1) \times 3^4 - 1 \\ &= 80q \times 3^4 + 3^4 - 1 \\ &= 80q \times 3^4 + 80 \end{aligned}$$

$$= 80(3^4 xq + 1)$$

$$= 80p$$

$$\text{where } p = 3^4 xq + 1$$

and p is an integer.

∴ if the statement is true for $n=k$, then
is also true for $n=k+1$.

∴ since true for $n=1$, also true for $n=1+1=2$,
and hence for all integers $n \geq 1$, by the
principle of Mathematical Induction.

Notes: a number of students tried to prove

$$3^{4k+1} - 1 = 80p \quad \text{instead of} \quad 3^{4(k+1)} - 1 = 80p$$

and ran into difficulties. Be very careful with
the initial algebra!

Q3

a)

i) $\angle COB = 70 \times 2 = 140^\circ$ $\left\{ \begin{array}{l} \angle \text{ at the centre is twice} \\ \text{the } \angle \text{ at the Circumf} \\ \text{Standing on the Same Arc} \end{array} \right.$

ii) $\angle OCB = \angle OBC$ (Base $\angle$'s of Isosceles Δ)

$$2 \times \angle OCB = 180^\circ - 140^\circ \quad (\text{Sum } \angle \text{'s of a } \Delta)$$

$$\angle OCB = \frac{40}{2} = 20^\circ \checkmark$$

★ All students did well in this question.

b) $4^n \geq 1 + 3n$ for $n \geq 1$

• Prove it is true for $n=1$

$$4^1 \geq 1 + 3 \quad \text{which is true. } \checkmark$$

• Assume it is true for $n=k$

$$4^k \geq 1 + 3k \quad \text{OR } \checkmark$$

$$4^k - 3k - 1 \geq 0 \quad (*)$$

• Prove it is true for $n=k+1$

★ There are several ways of proving this kind of question.

$$4^{k+1} - 3(k+1) - 1 \geq 0 \quad \checkmark$$

$$4^{k+1} - 3k - 4 \geq 0$$

$$4(4^k - 3k - 4) + 9k \geq 0 \quad \checkmark$$

This is true as

Some students did not show the proper and logical prove. It is true for $n=k+1$, but it is true for $n=1$. It is proven by Mathematical induction for $n \geq 1$.

2014 Ext 1 Task 1.

Q4 (a) $x = 2t - 1$ $y = 3t^2 + 1$

$$t = \frac{x+1}{2}$$

$$y = 3\left(\frac{x+1}{2}\right)^2 + 1$$

$$y = 3\left(\frac{x^2 + 2x + 1}{4}\right) + 1$$

Comment: Students forgot to square the denominator. Also some didn't "multiply the +1" by 4.

$$4y = 3x^2 + 6x + 7.$$

or

$$(x+1)^2 = \frac{4}{3}(y-1).$$

(b)(i) $P(6p, 3p^2)$ $Q(6q, 3q^2)$

$$m_{PQ} = \frac{3p^2 - 3q^2}{6p - 6q}$$

$$= \frac{1}{2}(p+q).$$

Comment: Many students used the point $C(8,0)$ instead of P and Q .

can

$$y - 3p^2 = \frac{1}{2}(p+q)(x - 6p).$$

$$y - 3p^2 = \frac{1}{2}(p+q)x - 3p^2 - 3pq$$

$$y = \frac{1}{2}(p+q)x - 3pq$$

(ii) $C(8,0)$

$$0 = \frac{1}{2}(p+q) \times 8 - 3pq$$

$$3pq = 4(p+q).$$

(iii)

$$M \left(\frac{6p+6q}{2}, \frac{3p^2+3q^2}{2} \right)$$

$$m \left(3(p+q), \frac{3}{2}(p^2+q^2) \right)$$

$$x = 3(p+q) \Rightarrow p+q = \frac{x}{3}$$

$$y = \frac{3}{2}(p^2+q^2)$$

$$y = \frac{3}{2}((p+q)^2 - 2pq)$$

$$\text{From part (ii)} \quad pq = \frac{4}{3}(p+q)$$

$$y = \frac{3}{2}((p+q)^2 - \frac{8}{3}(p+q))$$

$$y = \frac{3}{2} \left(\left(\frac{x}{3} \right)^2 - \frac{8}{3} \left(\frac{x}{3} \right) \right)$$

$$= \frac{3}{2} \left(\frac{x^2}{9} - \frac{8x}{3} \right)$$

$$y = \frac{x^2}{6} - \frac{8x}{6}$$

$$6y = x^2 - 8x \quad \text{or} \quad (x-4)^2 = 4 \left(\frac{1}{2} \right) \left(y + \frac{8}{3} \right)$$

5 a) i) $x^\circ = \hat{A}BP$ (alt $\angle$ s, $AC \parallel BP$)

$\hat{A}CB = \hat{A}BP$ ($\angle$ s in alt segments)

$\therefore x^\circ = \hat{A}CB$

$\therefore ABC$ is isosceles (base $\angle$ s equal)

$\therefore AB = BC$

ii) $\hat{A}CB = \hat{B}AP$ ($\angle$ s in alt segments)

$\therefore x^\circ = \hat{C}AB = \hat{B}CA = \hat{A}BP = \hat{B}AP$

$\hat{A}BC + 2x = 180^\circ$ ($\angle$ sum of $\triangle ABC$)

$\hat{APB} + 2x = 180^\circ$ (i.e. " " " " ABP)

$\therefore \hat{A}BC = \hat{APB}$

The angles are in the wrong position to use congruence.

b) $m_{OP} \times m_{OQ} = -1$

$$\frac{p^2 - 0}{2p - 0} \times \frac{q^2 - 0}{2q - 0} = -1$$

$$\frac{pq}{4} = -1$$

$\therefore pq = -4$

Finding the gradient of PQ does not help to answer the question

c) $\hat{E}BC + 95^\circ = 180^\circ$ (opp $\angle$ s of a cyclic quad)

$\hat{E}BC = 85^\circ$

$\hat{A}FE = \hat{E}BC$ (ext $\angle$ of a cyclic quad)

$= 85^\circ$

Question 6 (8 Marks)

a) Let $S(n): 1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + n \times 2^n = (n-1)2^{n+1} + 2$

For $n=1$:

$$\text{LHS} = n \times 2^n$$

$$= 1 \times 2^1$$

$$= 2$$

$$\text{RHS} = (n-1)2^{n+1} + 2$$

$$= (1-1)2^{1+1} + 2$$

$$= 2$$

✓

$\therefore \text{LHS} = \text{RHS} \therefore S(n)$ is true for $n=1$.

Assume $S(k)$ is true for some integer k , $k \geq 1$

$$\text{That is: } 1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + k \times 2^k = (k-1)2^{k+1} + 2$$

Prove $S(k+1)$ is true. That is show that:

$$1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + k \times 2^k + (k+1) \times 2^{k+1} = (k+1-1)2^{k+1+1} + 2$$
$$= k \cdot 2^{k+2} + 2 \quad \checkmark$$

$$\text{LHS} = 1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + k \times 2^k + (k+1)2^{k+1}$$

$$= \underbrace{(k-1)2^{k+1} + 2}_{\text{from assumption}} + (k+1)2^{k+1}$$

$$= 2^{k+1} (k-1 + k+1) + 2$$

$$= 2^{k+1} \cdot 2k + 2$$

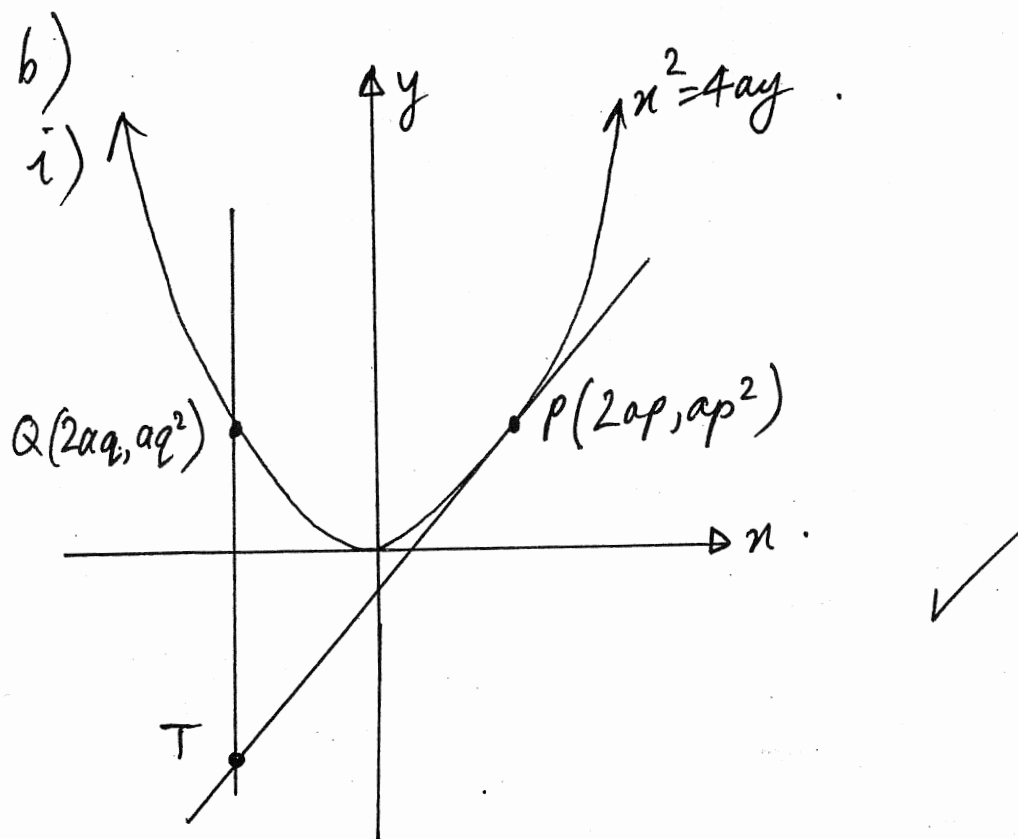
$$= k \cdot 2^1 \cdot 2^{k+1} + 2$$

$$= k \cdot 2^{k+2} + 2$$

$$= \text{RHS}$$

✓

$\therefore S(n)$ is true for $n=k+1$, whenever it is true for $n=k$. $S(n)$ is true for $n=1$ and by the Principle of mathematical induction is true for all integers $n \geq 1$.



ii) when $x = 2aq$: $y = p(2aq) - ap^2$
 $= 2apq - ap^2$

$\therefore T = (2aq, 2apq - ap^2)$

iii) Midpoint of $PT = \left(\frac{2ap + 2aq}{2}, \frac{2apq - ap^2 + ap^2}{2} \right)$
 $= \left(\frac{2a(p+q)}{2}, \frac{2apq}{2} \right)$
 $= (a(p+q), apq)$

Locus: $\begin{cases} x = a(p+q) \\ y = apq \end{cases}$

Since $pq = -1$: $y = -a$

$\therefore$ Locus is the directrix of the parabola $x^2 = 4ay$.

Question 7

- (a) (i) Let (x_1, y_1) be the external point.

$$a = 1$$

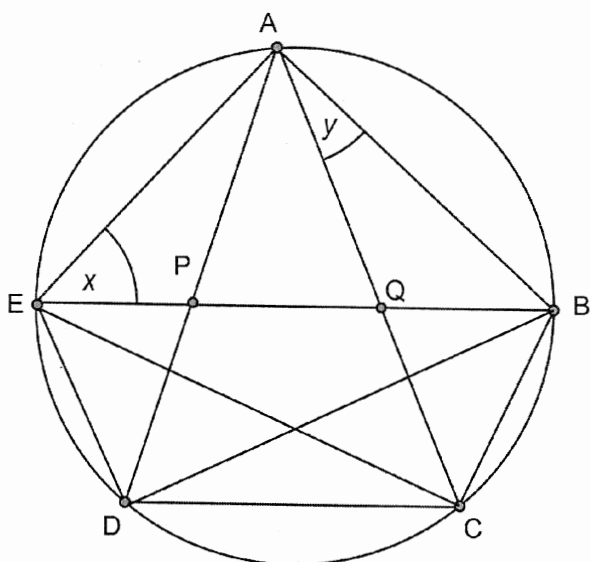
$$xx_1 = 2a(y + y_1) \quad \therefore xx_1 = 2(y + y_1)$$

$$y = \frac{x_1}{2}x - y_1 \quad \therefore \frac{x_1}{2} = 1, -y_1 = 3$$

Hence the external point is $(2, -3)$.

COMMENT :

Many students did not recognise the opportunity to use the chord of contact formula and spent much longer on the solution than was required.



COMMENT :

Many students did not attempt this question fully – presumably due to time running out at the end of the exam period.

As with all proofs, please ensure reasoning is provided for each step of your explanation.

- (b) (i) $\triangle BAE$ is isosceles (given, $AB = AE$)

$$\angle ABE = \angle AEB = x \text{ (base } \angle\text{s of isos. } \triangle)$$

$$\angle ABE = \angle ECA \text{ (} \angle\text{s in same segment)}$$

$$\therefore \angle AEB = \angle ECA = x \text{ (both equal } \angle ABE)$$

- (ii) $\angle ABE = x$ (proven above)

$$\angle BQC = \angle ABQ + \angle BAQ \text{ (ext. } \angle \text{ of } \triangle AQB)$$

$$\therefore \angle BQC = x + y$$

- (iii) $\angle BDC = \angle BAC = y$ ($\angle$ s in same segment)

$$\angle ADB = \angle AEB = x \text{ (} \angle\text{s in same segment)}$$

$$\therefore \angle PDC = \angle ADB + \angle BDC = x + y$$

$$\therefore PQCD \text{ is a cyclic quadrilateral}$$

$$\text{(ext. } \angle \text{ of quad. is equal to opp. int. } \angle)$$